

Wudneh Tilahun Mengist

Mathematics Lecturer & Researcher

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PROFESSIONAL PROFILE

I am a dedicated Mathematics Lecturer and Researcher based in the Department of Mathematics at Debre Tabor University, Ethiopia.

Holding a Master of Science degree in Mathematics specializing in Numerical Analysis from Bahir Dar University and a Bachelor of Science in Applied Mathematics from Arba Minch University, my academic foundation focuses on solving complex mathematical models efficiently.

Since 2011, I have been deeply committed to cultivating undergraduate excellence across fundamental and applied courses including Numerical Analysis, Calculus, and Number Theory.

Beyond traditional instruction, I am an active proponent of modern instructional design frameworks and open educational resources.

My technical expertise spans advanced mathematical software, data analysis, and professional typesetting tools such as LaTeX, Python, MATLAB, and Wolfram Mathematica.

Whether developing open-access digital learning modules on Open edX or curating educational content for my channel, I strive to make mathematics intuitive and widely accessible.

EDUCATION

Master of Science (MSc) - Bahir Dar University (2008 – 2010)

Specialization: Numerical Analysis

Bachelor of Science (BSc) - Arba Minch University (2003 – 2007)

Specialization: Applied Mathematics

PROFESSIONAL EXPERIENCE

Lecturer in Mathematics - Debre Tabor University (2011 – Present)

Undergraduate Teaching & Research (2011–Present)

Creator & Educator - The Hybrid Math Hub (2020 – Present)

Digital Content Production & Blended Learning

Layout Editor - Journal of Interdisciplinary Science & Technology (JIST) (2023 – Present)

Typesetting, Design, and Editorial Standards

Master / Field Trainer - National Election Board of Ethiopia (2021 – 2026)

Operational Training & Leadership (2021–2026)

PUBLICATIONS & RESEARCH

An exploration of quintic Hermite splines to solve Burgers' equation

Arabian Journal of Mathematics (Springer Nature) (2019) - PUBLISHED

Metrics: 1212 accesses, 13 citations

An Adaptive B-Spline Collocation Framework for Singularly Perturbed Boundary Value Problems with Boun

Journal of Interdisciplinary Science and Technology (JIST) (2025) - REVIEW

Teacher Capacity Building Framework for Mathematics Educators

Regional Training Initiative (2024) - PROJECT

CERTIFICATIONS & TRAINING

Financial Services & Capital Markets - Ethiopian Securities Exchange (ESX) Digital Academy

Completed professional training modules covering modern financial ecosystems, market operations, and investment structures.

Interactive Digital Learning & Teaching (IDLT) - Debre Tabor University

Served as an IDLT Trainer, facilitating courses and mentoring faculty on modern digital learning frameworks and Open edX integration.

Master & Field Trainer (Electoral Operations) - National Election Board of Ethiopia

Certified and deployed across regional training cycles as an official Master and Field Trainer (2021-2026).

English Language Improvement Program (ELIP) - DTU Academic Development

Completed specialized professional development focused on advanced academic communication, instructional delivery, and pedagogy enhancement.

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